

Egor Petrov

Researcher

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| [GitHub: modernTalker](#)

| [Google Scholar](#)

Education

Moscow Institute of Physics and Technology (MIPT)

Moscow, 2022 – 2026 (Expected)

B.S. in Applied Mathematics and Computer Science.

Data Science Track (MIPT & Yandex School of Data Analysis)

Moscow, 2024 – 2026

Specialized curriculum in Data Science and Statistics.

Research & Engineering Experience

Yandex Research

Jul 2025 – Present

Research Resident, Moscow

- Research on **asynchronous optimization**.
- Theoretical research on **one-step delayed optimization**, enhancing convergence proofs.

BRAIn Laboratory, MIRIAI

Feb 2024 – Present

Researcher, Moscow

- Pioneered memory-efficient zeroth-order optimization methods for fine-tuning, achieving a **50% reduction in memory footprint**.
- Developed and theoretically analyzed novel **stochastic and zeroth-order algorithms** for optimization, validated on variety of domains.
- Developed a novel Hessian-based approach for **Learning Theory**.

Yandex

Aug 2024 – Jan 2025

Developer, Personalization Quality R&D Group, Moscow

- Engineered a production pipeline integrating Vowpal Wabbit-based online features into a ranking model, achieving a **0.1% uplift in core ranking metrics** (AUC, nDCG).

Publications

- **Leveraging Coordinate Momentum in SignSGD and Muon: Memory-Optimized Zero-Order..**
E. Petrov, G. Evseev, A. Antonov, A. Veprikov, P. Plyusnin, N. Bushkov, et al.
ICML Workshop on Tiny Titans, 2025. [\[Paper\]](#)
- **Sign-SGD is the Golden Gate between Multi-Node to Single-Node Learning...**
D. Medyakov, S. Stanko, G. Molodtsov, P. Zmushko, G. Evseev, E. Petrov, et al.
ICLR, 2026 [\[Paper\]](#)
- **When Extragradient Meets PAGE: Bridging Two Giants to Boost Variational Inequalities**
G. Molodtsov, V. Parfenov, E. Petrov, E. Grigoriy, D. Medyakov, A. Beznosikov.
UAI, 2025. [\[Paper\]](#)
- **Shuffling Heuristic in Variational Inequalities: Establishing New Convergence Guarantees**
D. Medyakov, G. Molodtsov, E. Grigoriy, E. Petrov, A. Beznosikov.
ICOMP 2024. Submitted to Q1 Journal JOTA. Awarded **1st place** at Neuroinformatics 2024. [\[Paper\]](#)
- **Zero Order Algorithm for Decentralized Optimization Problems**
A. Veprikov, E. Petrov, G. Evseev, A. Beznosikov.
Published in Doklady Mathematics, 2024 [\[Paper\]](#)
- **Sampling of Semi-Orthogonal Matrices for the Muon Algorithm**
E. D. Petrov, G. V. Evseev, A. V. Antonov, A. S. Veprikov, N. A. Bushkov, S. V. Moiseev, A. N. Beznosikov.
Published in Doklady Mathematics, 2025 [\[Paper\]](#)
- **Closing the Curvature Gap: Full Transformer Hessians**
Egor Petrov, Nikita Kiselev, Vladislav Meshkov, Andrey Grabovoy.
Submitted to COLT 2026. [\[Paper\]](#)

Teaching Experience

- **Central University** – Teaching Assistant [2025 – Present]
Serve as a teaching assistant for the *Scientific Studio* and *Python for Data Analysis*; responsibilities include mentoring scientific projects and checking homework assignments.
- **Moscow Institute of Physics and Technology** – Teaching Assistant [2024 – 2025]
Served as a teaching assistant for the *Algorithms and Data Structures* course; responsibilities included mentoring students' progress and providing consultations.